



# Mission-Adaptive Lifting System Design using Integrated Multidisciplinary Topology Optimization

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A coupled aero-structural optimization framework for the design of a mission adaptive airfoil is presented. A two-dimensional linear finite element solver and a constant strength doublet flow solver, along with a load-transfer and mesh displacement scheme constitute the fluid-structure interaction system. The morphing airfoil configuration is synthesized by concurrent consideration of both aerodynamic and structural design objectives. A topology optimization sub-problem is solved for the conceptual design of an internal compliant mechanism that morphs the airfoil shape based on different  $c_l$  requirements. A three-dimensional flow-field analysis using high-fidelity RANS-based simulations indicates that the mission adaptive airfoil configurations outperform a baseline NACA 0012 airfoil at all design points. Instantaneous flow-field characterization in the trailing edge region shows the evolution of distinct asymmetric flow features as the design  $c_l$  increases.

## I. Nomenclature

|                |   |                                          |
|----------------|---|------------------------------------------|
| $\mathbf{A}$   | = | aerodynamic residual                     |
| $\mathbf{AIC}$ | = | aerodynamic influence coefficient matrix |
| $c$            | = | chord length                             |
| $c_d$          | = | drag coefficient                         |
| $c_l$          | = | lift coefficient                         |
| $c_l^*$        | = | optimized lift coefficient               |
| $f$            | = | objective function                       |
| $\mathbf{F}^a$ | = | vector of aerodynamic forces             |
| $F_{in}$       | = | force at input port                      |
| $\mathcal{G}$  | = | coupled aeroelastic residual             |
| $\mathbf{I}$   | = | identity matrix                          |
| $\mathcal{J}$  | = | global objective function                |
| $\mathbf{K}$   | = | elastic stiffness matrix                 |
| $\mathbf{L}$   | = | basis vector                             |
| $m$            | = | mass                                     |
| $m_{max}$      | = | maximum mass                             |
| $n_{struct}$   | = | number of structure design variables     |
| $p$            | = | penalization parameter                   |
| $\mathcal{S}$  | = | structure residual                       |
| $\mathbf{T}_1$ | = | displacement transfer matrix             |
| $\mathbf{T}_2$ | = | load transfer matrix                     |
| $\mathbf{U}$   | = | structural displacement vector           |
| $U_{out}$      | = | structural displacement at output port   |
| $V$            | = | flight speed                             |
| $\mathcal{X}$  | = | global design variables                  |

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|           |   |                        |
|-----------|---|------------------------|
| $\alpha$  | = | angle of attack        |
| $\beta$   | = | panel angle of attack  |
| $\Delta$  | = | gradient               |
| $\lambda$ | = | coefficient vector     |
| $\mu$     | = | doublet strength       |
| $\phi$    | = | interpolation function |
| $\rho$    | = | material density       |
| $\theta$  | = | rotation angle         |
| $\xi$     | = | OML interface          |
| $\zeta$   | = | FFD control point      |

## II. Introduction

IN the current day and age, aircraft design is generally governed by a multitude of factors such as aerodynamic efficiency, carbon footprint, overall cost of manufacturing and operational sustainability. Recent trends in the global aviation market suggest a shift from large four-engine multi-aisle aircraft to smaller twin engine aircraft configurations [1]. In conservative terms, the design of such aircraft systems is primarily governed by efficient sub-systems such as controls, structures, guidance, propulsion and aerodynamics. Holistic design consideration of these sub-systems contributes towards the economic success of the aircraft in a competitive global aviation market [2].

An aircraft configuration is a collection of complex sub-systems and the quest to achieve an efficient design configuration has been a never-ending quest in the aerospace community. A multitude of studies in the past have used both experimental and numerical optimization techniques to achieve drag-optimal wing design configurations. Studies such as those by Drela [3] and Samareh [4] discuss the application of gradient-based optimization routines using panel-based methods to obtain shape-optimized airfoil designs. More evolved approaches to aerodynamic shape optimization, especially for planar and non-planar wing configurations have been reported by Ranjan *et. al* [5], Ranjan *et. al* [6], Martins *et. al* [2] and Palacios *et. al* [7]. Multi-disciplinary design studies such as those by James *et. al* [8], Kennedy *et. al* [9] and Elham *et. al* [10], approach wing design from an integrated aero-structural perspective wherein both aerodynamic and structural design parameters constitute the optimization formulation. While the aforementioned studies discuss novel approaches to wing design, a key limiting factor of these design approaches is that the resulting optimal design configurations are optimized only with respect to a single flight condition. An aircraft design synthesis usually begins with a clear definition of mission objectives and corresponding performance deliverables. For instance, a general aviation aircraft configuration usually operates on a flight trajectory consisting of flight phases such as take-off, climb, cruise, maneuver, descent, approach and landing, each with different overall  $L/D$  requirements. Such multi-point design considerations during wing optimization has been widely reported in the open literature. Key seminal studies include those by Drela [3], Nemec *et. al* [11] and Kenway *et. al* [12]. A multi-point optimal airfoil or wing design outperforms a single point optimal configuration, when the full mission is considered. However the gains in aerodynamic performance parameters such as overall  $L/D$  are more evenly distributed over the operating flight envelope, thus limiting the true performance potential of the design at each design point. Such performance trends are often a result of weighted optimization techniques, as discussed by Nemec *et. al* [11]. Nevertheless, multi-point design approaches are an effective means to achieve design configurations that have optimal performance index over a wider range of flight regimes [11].

An alternative design approach for optimal lifting configurations is to use morphing or mission-adaptive configurations. An adaptive lifting system, as defined by Weisshear [13], is a wing which has the ability to actively modify wing-section camber, local incidence or span-wise camber, while maintaining a seamless outer mould line (hereafter OML). This enables the wing to produce near-optimal configurations for each individual flight regime discussed earlier. Some of the key features of a mission adaptive wing include the ability to achieve variable camber, roll control, gust load alleviation, maneuver load control and direct lift mode. Significant advantages of the mission adaptive wing concept were first demonstrated by a series of experimental flight test campaigns by NASA, summarized by Gilbert [14] and Bonnema *et. al* [15]. Both studies reported the largest improvements due to smooth and continuous variations of the OML, especially at the off-design conditions where the mission adaptive configuration was able to reconfigure itself to meet design requirements at that point. However, the complex nature of the actuation system, and the added weight penalty due to necessary internal mechanisms mitigated the practical application of the the adaptive wing concept.

Numerous studies in the past three decades have reported improvements in the actuation technology such as the direct or integrated use of active materials into compact hybrid actuators [16–18]. Such recent developments in actuation technology have revived the interest of the aerospace community in the mission adaptive wing concept. Much of the

recent work done towards morphing structures can be broadly classified into two groups [19] : 1) Multiply redundant control surfaces [20] and 2) Conformal shape changes [21]. The former class of studies explore the use of traditional control surfaces to change the OML. Such systems are often plagued with increased weight and aerodynamic drag. The second class of studies achieve a smooth and seamless OML profile using distributed actuation architecture working in conjunction with kinematic mechanisms. the Use of active materials to achieve conformal shape changes have also been explored, however, due to their limited force and stroke generation capabilities, their application has proven to be less effective [20].

A mission adaptive wing section consists of an OML skin on which aerodynamic pressure forces are applied, an internal kinematic mechanism that transfers the load from the OML to the actuator and vice-versa, and an internal support structure that distributes the forces to other wing section stations as well as the fuselage. Hence, any process to synthesize an adaptive wing section much include these key elements and must take into account the coupled aero-elastic response of the lifting system. Since the aerodynamic forces acting on the OML are dependent on the structural deformations and vice-versa, concurrent aero-structural design considerations are necessary for the design of an adaptive wing section, thus suggesting the need for a multi-disciplinary design optimization framework. Studies such as those by Maute *et.al* [19] and Maute *et. al* [22], to name a few, have demonstrated the application of a coupled aero-structural design optimization approach towards the development of compliant/mission adaptive wings and wing sections.

The current study presents a multi-disciplinary design optimization framework that results in the synthesis of a mission adaptive airfoil. A coupled aero-structural system is presented that forms a part of a larger optimization framework. Considering the key components that constitute a mission adaptive airfoil, different numerical techniques are implemented for each discipline and are coupled to form a monolithic architecture. For instance, a constant strength doublet lattice routine is implemented to approximate flow-field over thick airfoils. This allows both thickness and camber parameters to be treated as design variables, as opposed to a lumped vortex method wherein only camber variations can be treated as design variables [23]. For the internal mechanism synthesis, a linear elasticity-based topology optimization routine is implemented. Due to the large design space in the coupled aero-structural domain, an adjoint sensitivity formulation is used to obtain the analytical derivatives. The use of a coupled doublet lattice method in the subsonic flight regime for the development of mission adaptive configurations is a novel contribution of this work. Another key point of novelty is that the resulting internal mechanisms are physically realizable.

### III. Integrated design framework

The key components that constitute the large optimization framework needed to synthesize an adaptive airfoil configuration are illustrated in Fig. 1, and are considered as the cornerstones of the current study. The following sections briefly describe some of the sub-modules critical to the current study.



**Fig. 1 Sub-modules of the integrated design optimization framework**

### A. Shape parameterization

The Free-Form Deformation ( hereafter FFD ) method [24] is chosen as the shape parameterization method in the current study. The FFD is a robust, accurate and versatile method for both local and global changes wherein a native shape profile is embedded within a flexible control box. An appropriate mathematical mapping relates the physical space on the native shape profile to the parametric space defined on the flexible control box. By deforming these control points in the parameter space, changes to the interior physical space are obtained in a smooth and consistent manner. B-spline basis functions that are used for mapping deformations from a physical space to the parametric space, are defined as:

$$B(t) = \sum_{i=0}^n \binom{n}{i} (1-t)^{n-i} t^i \zeta_i \quad (1)$$

where  $\zeta_i$  are the equidistant control points. A linear system for  $\zeta_i$  is built using the native airfoil configuration OML coordinates and the desired number of control points. For the design of a feasible Mission Adaptive Airfoil ( hereafter MAA ) configuration, the conventional FFD framework is modified to include rotation angle of the airfoil coordinates as a design variable. This allows us to achieve a design space wherein the airfoil OML can be rotated by an angle  $\theta$ , about a specified rotation point. The un-deformed control box is modeled using linear functions:

$$y_{up} = \frac{x \times \sin(-\theta) + 0.1}{\cos(\theta)} \quad (2)$$

and

$$y_{down} = \frac{x \times \sin(-\theta) - 0.1}{\cos(\theta)} \quad (3)$$

These modifications to the FFD shape parameterization framework allows the airfoil to deform without altering the local geometry of the trailing edge ( hereafter T.E ). Hence, problems such as T.E cross-over during optimization which may cause the flow solver to diverge, are eliminated.

### B. Constant Strength Doublet Flow Solver

A constant strength doublet flow solver is chosen as the linear potential flow solver for the current study. The main motivation behind choosing a doublet-based method over a discrete vortex method is because of its ability to compute satisfactory surface pressure distributions for thick lifting airfoils. This allows for inclusion of airfoil thickness in addition to airfoil camber, as opposed to only camber considerations in the case of a lumped vortex method. The underlying numerical framework is described elsewhere [23].

### C. Topology Optimization

Topology optimization provides a means for achieving optimal structural designs wherein the shape, sizing and topology of the members are optimized, subject to external loading. Topology optimization as a material distribution problem was first proposed by Bendsoe and Kikuchi [25]. The design domain is discretized using finite elements, which leads to an ill-posed integer optimization formulation. Each finite element is assigned a non-physical density variable and the Solid Isotropic Material Penalization Method (SIMP) is used to express the effective young's modulus of each element as the product of the Young's modulus of the design material,  $E_0$ , and an interpolating function for the material density  $\rho_e$  [8]:

$$E_e = \phi(\rho_e) E_0 \quad (4)$$

The interpolating function  $\phi$  is chosen such that density at each finite element cell  $\rho_e$  is forced towards either a fully solid ( $\rho = 1$ ) or a fully void ( $\rho = 0$ ) phase. Based on the study by Bendsoe *et. al* [25],  $\phi$  can be expressed as:

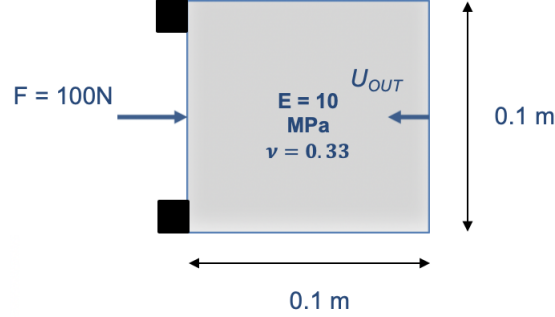
$$\phi(\rho_e) = \rho_e^p \quad (5)$$

hence

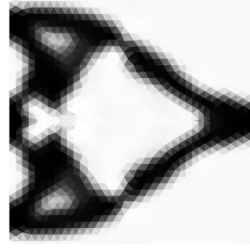
$$E_e = \rho_e^p E_0 \quad (6)$$

where, the penalization parameter  $p$  usually takes a value greater than 1. For designing a compliant mechanism using topology optimization, the objective of the problem formulation is to determine the mechanism that either minimizes or

maximizes the displacement at an output location, subject to an input displacement or a force. Fig. 2 illustrates the well-known displacement inverter compliant mechanism design problem. A force  $F_{in}$  is applied at the input port and a displacement  $U_{out}$  is desired at the output port. Fig. 2(b) shows the resulting mechanism that displaces the output port in a direction opposite to that of the applied force.



(a) Design domain and boundary conditions



(b) Compliant mechanism

**Fig. 2 Displacement inverter mechanism**

In this compliant mechanism design problem, the external load distribution is known a priori. Design of compliant mechanisms subject to aerodynamic loading must take into account the change in aerodynamic loading as the structure is displaced during optimization. Hence, an aeroelastic system is necessary in order to account for such coupled behavior.

#### D. Coupled Fluid-Structure Interaction Analysis

The discretized form of the steady state response of the aeroelastic system is expressed as:

$$\mathcal{G} = \begin{bmatrix} \mathbb{S}(U, \mu, \alpha) \\ \mathbb{A}(U, \mu, \alpha) \end{bmatrix} \quad (7)$$

where,  $\mathbb{S}$  represents the structural residual, expressed as:

$$\mathbb{S} = \mathbf{K}U - F_S(\mu, \alpha) = 0 \quad (8)$$

$\mathbb{A}$  represents the aerodynamic residual, expressed as:

$$\mathbb{A} = \mathbf{AIC} \cdot \mu - V = 0 \quad (9)$$

Here,  $U$  is the vector of nodal displacements,  $\mu$  is the doublet strength of each panel on the OML,  $\alpha$  is the global angle of attack,  $F^a$  is a vector of aerodynamic pressure forces acting at the fluid-structure interface ( in this case, the OML).  $\mathbf{AIC}$  is the aerodynamic stiffness matrix or matrix of aerodynamic influence coefficients, and  $\mathbf{K}$  is the structural stiffness matrix. If  $n_{struct}$  denotes the total structural degrees of freedom, then  $\mathbf{K}$  is an  $n_{struct} \times n_{struct}$  symmetric positive definite matrix. For the mesh motion solver, the total number of degrees of freedom is  $2 \times R$ , where  $R$  is the total number of grid points on the OML. A load and displacement transfer mechanism proposed by Brown [26] is used. For small elastic deformations, the aerodynamic system can be expressed as:

$$\mathbf{AIC}.\mu - V(\alpha + \beta_0) = 0 \quad (10)$$

where,  $V$  is the flow velocity and  $\beta_0$  is the local panel angle of attack

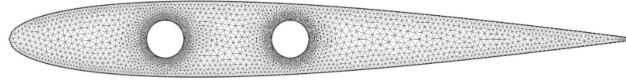
Next we introduce a transformation matrix  $T_1$  that relates the local angle of attack of each OML panel to the structural displacements and hence allows a full set of structural DOFs to be included in the equations:

$$\mathbf{AIC}.\mu - VT_1(U + U_0) = 0 \quad (11)$$

The aerodynamic forces generated by the panels cause an equivalent set of applied forces on the structural nodes. A transformation matrix  $T_2$  that maps the forces between the structural and aerodynamic nodes is defined as:

$$F_S = T_2 F^a \quad (12)$$

where,  $F_S$  is the vector of forces at OML structure nodes and  $F^a$  is the vector of aerodynamic pressure forces. Fig.3 depicts the Finite element mesh domain consisting of Delaunay tessellations. A total of 5411 constant-strain triangles make up the unstructured mesh. The mesh framework was developed based on the Delaunay-based triangulation scheme together with a Steiner point generation algorithm for defining interior nodes and a “breadth-first” search algorithm for mesh refinement [27]. One key advantage of this mesh generation approach is the exact allocation of grid nodes at each coordinate point on the OML, thus making the panel coordinates exactly aligned with the structural nodes.



**Fig. 3 Finite element mesh with Delaunay tessellations**

The two blocks of the coupled non-linear system of Eq.7 have distinct numerical properties, each with different optimal solution algorithms. A partitioned procedure for solving the non-linear system of equations is used wherein a Newton-based approach is employed for building linear sub-problems and a block Gauss-Seidel scheme is used to solve these sub-problems [19].

Linearizing the non-linear system of equations about the coupled system's state variables  $(U, \mu)$  yields:

$$\begin{bmatrix} \Delta \mathbb{S} \\ \Delta \mathbb{A} \end{bmatrix} = \begin{bmatrix} \frac{\partial \mathbb{S}}{\partial U} & \frac{\partial \mathbb{S}}{\partial \mu} \\ \frac{\partial \mathbb{A}}{\partial U} & \frac{\partial \mathbb{A}}{\partial \mu} \end{bmatrix} \begin{bmatrix} \Delta U \\ \Delta \mu \end{bmatrix} \quad (13)$$

The linearized tangent operator simplifies to :

$$\begin{bmatrix} \mathbf{K} & -\frac{\partial F^a}{\partial \mu} \\ \frac{\partial \mathbb{A}}{\partial U} & \mathbf{AIC} \end{bmatrix} \quad (14)$$

The residual  $\mathbb{S}$  and  $\mathbb{A}$  contain the structural and aerodynamic stiffness matrices respectively, and are solved using a direct method. The off-diagonal entries account for the coupling between the two disciplines. For instance,  $\frac{\partial \mathbb{A}}{\partial U}$  accounts for the effect of structural displacements on the aerodynamic system and is computed by evaluating the aerodynamic system residual OML displacements. The term  $\frac{\partial F^a}{\partial \mu}$  accounts for the effect of the doublet strengths on the aerodynamic force and is computed analytically as:

$$\frac{\partial F_{y_i}}{\partial \mu_i} = \frac{1}{2} \rho V^2 (-2 \cos(\beta_i - \alpha) \left[ \frac{2}{S_i + S_{i+1}} \right]^2) \quad (15)$$

$$(\mu_i + (X_{mid_i} - X_{mid_{i+1}}) \cos(\alpha) + (Y_{mid_i} - Y_{mid_{i+1}}) \sin(\alpha) - \mu_{i+1})$$

$$\frac{\partial F_{x_i}}{\partial \mu_i} = \frac{1}{2} \rho V^2 (-2 \sin(\beta_i - \alpha) \left[ \frac{2}{S_i + S_{i+1}} \right]^2) \quad (16)$$

$$(\mu_i + (X_{mid_i} - X_{mid_{i+1}}) \cos(\alpha) + (Y_{mid_i} - Y_{mid_{i+1}}) \sin(\alpha) - \mu_{i+1})$$

where,  $\beta$  is the local panel angle of attack,  $\alpha$  is the airfoil angle of attack,  $S$  is the panel span,  $\mu$  is the double strength at each panel and  $X_{mid}$  and  $Y_{mid}$  are the x and y coordinates for the mid-point for each panel. The analytical derivatives presented above are verified using finite differences. The system described in Eq.13 is solved iteratively for the states  $\mu$  and  $U$  at equilibrium using the block-Gauss-Seidel method.

#### IV. Coupled Aero-structural Sensitivities

A coupled aero-structural design optimization problem is high dimensional due to the large number of aerodynamic and structural design variables. Accurate and efficient computation of the coupled sensitivities is crucial for a gradient-based optimization algorithm since it drives the optimizer towards a local or a global solution in the large design space. In the current study, analytical sensitivities for the aero-structural design problem are calculated.

Consider the two governing systems of equation in Eqs. 8 and 9. Splitting the residual,  $\mathbb{R}$ , adjoint vector,  $\mathbb{V}$ , and state variable,  $\mathbb{Y}$ , yields:

$$\mathbb{R} = \begin{bmatrix} \mathbb{A} \\ \mathbb{S} \end{bmatrix}; \mathbb{V} = \begin{bmatrix} \lambda_S \\ \lambda_A \end{bmatrix}; \mathbb{Y} = \begin{bmatrix} \mu \\ U \end{bmatrix}; \quad (17)$$

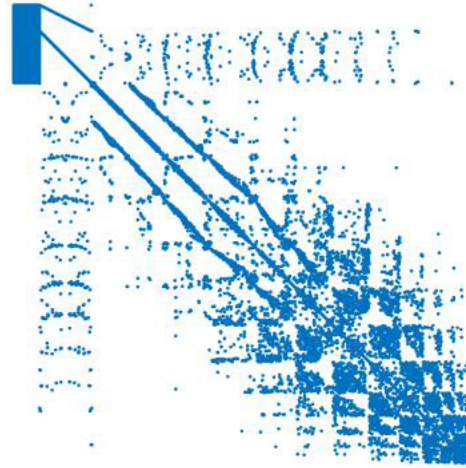
For an arbitrary single discipline, the adjoint equation is expressed as:

$$\frac{\partial \mathbb{R}}{\partial \mathbb{Y}} \mathbb{V} = -\frac{\partial f}{\partial \mathbb{Y}} \quad (18)$$

where  $f$  is a chosen function of interest. Expanding Eq. 18 to an aero-structural system in discrete adjoint form:

$$\begin{bmatrix} \frac{\partial \mathbb{S}}{\partial U} & \frac{\partial \mathbb{S}}{\partial \mu} \\ \frac{\partial \mathbb{A}}{\partial \mu} & \frac{\partial \mathbb{A}}{\partial U} \end{bmatrix}^T \begin{bmatrix} \lambda_S \\ \lambda_A \end{bmatrix} = - \begin{bmatrix} \frac{\partial U_{out}}{\partial U} \\ \frac{\partial c_l}{\partial \mu} \end{bmatrix} \quad (19)$$

The tangent matrix in Eq. 19 is the same as the tangent matrix from Eq. 13 and its sparsity pattern is shown in Fig. 4. The column vectors on the right-hand size are obtained from the adjoint solutions of the structural and aerodynamic discipline. Eq. 19 is solved for  $\lambda_S$  and  $\lambda_A$ , the structural and aerodynamic adjoint vectors, respectively.



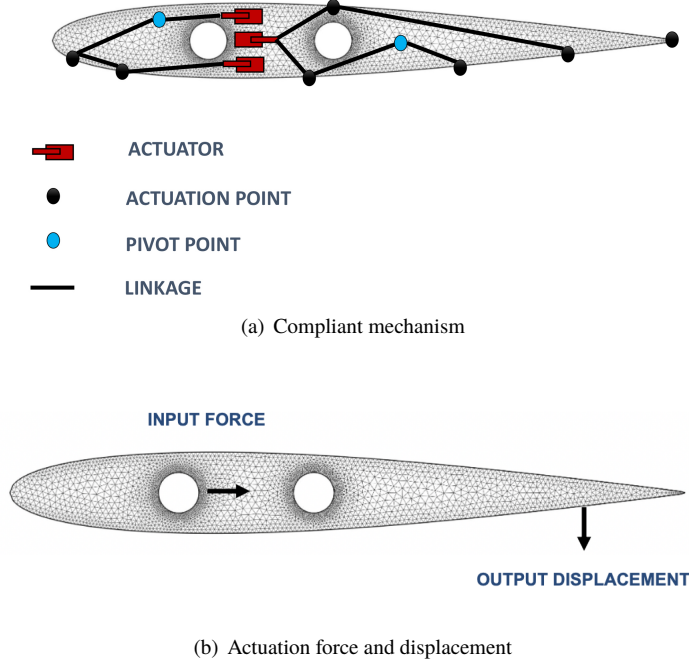
**Fig. 4 Sparse tangent matrix**

The total sensitivity of the objective function  $\mathcal{J}$  with respect to global design variables  $\mathcal{X}$ , is written as:

$$\frac{d \mathcal{J}}{d \mathcal{X}} = \frac{\partial \mathcal{J}}{\partial \mathcal{X}} + \lambda_A \frac{\partial \mathbb{A}}{\partial \mu} + \lambda_S \frac{\partial \mathbb{S}}{\partial U} \quad (20)$$

## V. Problem formulation

An adaptive wing section can be designed using either of two ways: a sequential design approach wherein the OML is optimized for increased aerodynamic performance followed by structural optimization using a fixed optimal OML from the previous step. Such a two-step design approach often results in diminished performance as the structural optimizer tries unsuccessfully to match the aerodynamically optimal OML profile [19]. The current study focuses on a multidisciplinary design approach wherein both the aerodynamic and structural design criteria are considered concurrently in the form of objective and design constraints.

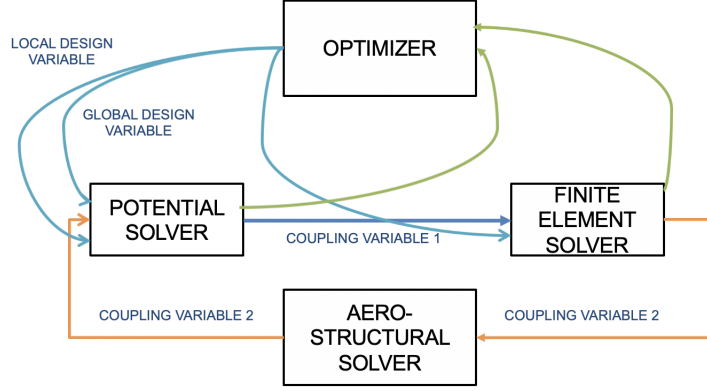


**Fig. 5 Conceptual internal mechanism**

Fig. 5(a) shows the conceptual design for an internal kinematic mechanism for the adaptive airfoil design problem. The internal mechanism transfers the actuation forces to the flexible OML and transfers back the external pressure forces to the actuator and internal support structure. The shape, size and location of the design domain in which the internal support structure will be placed, are known a priori as the design profile of these structures are driven by overall wing-section structural stiffness requirements [19]. In the current study, the boundary of the two spars (i.e. the two circles near the middle of the airfoil domain) are taken as the internal supports, where the degrees of freedom are fixed. Fig. 5(b) illustrates location and direction of the input actuator force as well as the output port where displacement is desired.

The multi-disciplinary framework is illustrated in Fig. 6. The optimization formulation for the design of a mission adaptive airfoil is expressed as:





**Fig. 6 Integrated design optimization flow chart**

$$\begin{aligned}
 &\text{minimize} && U_{out} \\
 &\rho, \zeta \\
 &\text{subject to} \\
 &\mathbf{K}U = F \\
 &\mathbf{AIC}\mu = V \\
 &c_l - \tilde{c}_l \leq 0 \\
 &\frac{M(x)}{M_0} - \tilde{M} \leq 0 \\
 &C - \tilde{C} \leq 0 \\
 &\rho_{min} \leq \rho \leq 1
 \end{aligned} \tag{21}$$

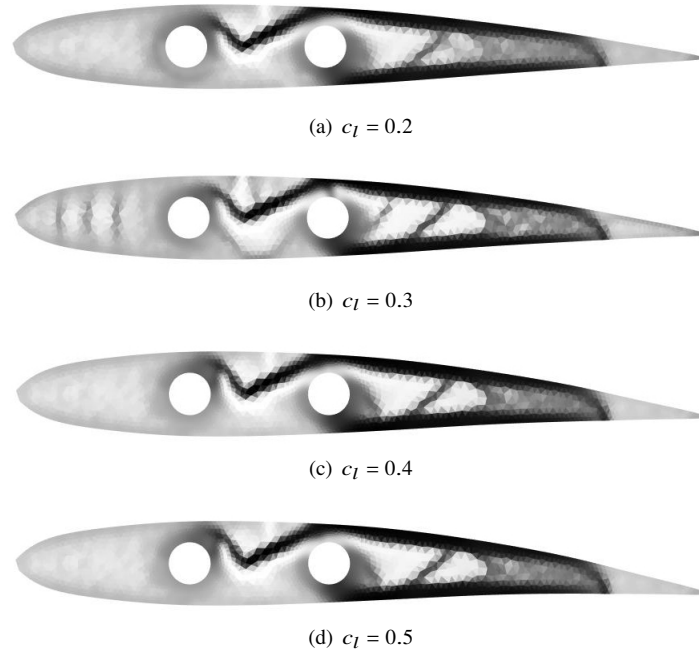
In the above problem formulation, the optimization algorithm minimizes the displacement at a specified output port which in turn changes the camber of the OML and hence the aerodynamic characteristics. An actuation force of 500 N is applied at  $x/c = 0.32$  and  $y/c = 0$  to enable vertical displacement at the output port located at  $x/c = 0.85$ , lying on the pressure side of the OML. The actuation force is defined in the force vector  $F$  of the first constraint. The third constraint enforces a desired  $c_l$  which in turn defines the optimal OML profile. The fourth constraint limits the amount of material used for the design of the internal mechanism. The fifth constraint limits the elastic compliance of the structure, thereby ensuring some requisite stiffness, and encouraging connected design with a contiguous material distribution from the actuator location (input port), to the output port. The bounding limit on compliance is set to its value at the end of first optimization iteration. Therefore, we constrain the structural stiffness so that the optimized structure remains as least as stiff as the baseline initial structure. The last constraint is a box constraint for the element densities to avoid any singularities in the finite element calculation. Here, the minimum value of element density is  $\rho_{min} = 0.01$ .

## VI. Results and discussion

A NACA 0012 airfoil, at aero-elastic equilibrium, is chosen as the baseline configuration for the current study. Inviscid flow conditions are implemented at a subsonic Mach number to avoid any non-linear compressibility effects. Table VI lists all the material, geometric and flow properties used in the current study.

| Material properties        |                   |
|----------------------------|-------------------|
| E                          | 50 MPa            |
| $\nu$                      | 0.33              |
| Geometric properties       |                   |
| c                          | 1 m               |
| spar 1                     | $0.25 \times x/c$ |
| spar 2                     | $0.45 \times x/c$ |
| spar radius                | $0.03 \times x/c$ |
| Flow properties : Solution |                   |
| Ma                         | 0.25              |
| Flow properties : Analysis |                   |
| Reynolds number            | $6 \times 10^6$   |

As a first case, the baseline NACA 0012 airfoil is subject to flow at Ma 0.25 and angle of attack  $\alpha = 0^\circ$ . An adaptive configuration is desired such that the airfoil  $c_l$  is increased from 0 to 0.5 while keeping the angle of attack  $\alpha$  and Mach number constant under aero-elastic equilibrium conditions. Using the optimization formulation from Eq.21, an internal compliant mechanism that deforms the shape to produce these desired  $c_l$  are achieved and are shown in Fig. 7.

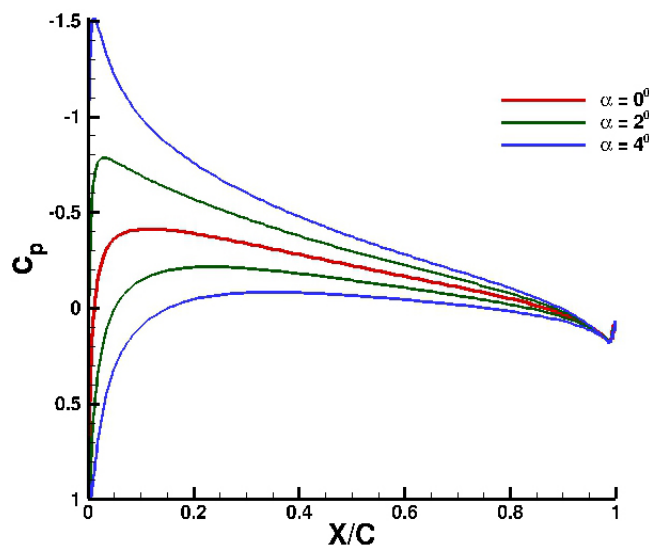


**Fig. 7 Adaptive airfoil configuration at  $\alpha = 0$**

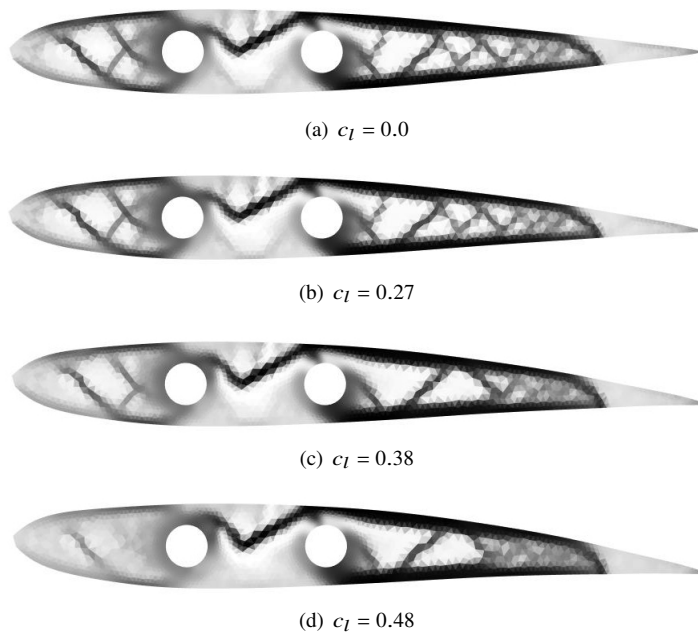
A seamless OML profile with a continuous material distribution for the internal kinematic mechanism is achieved. While the compliant mechanism itself is well-defined within the structure, intermediate material distributions for both aft and fore regions are observed. Although close observation of Fig.7(b) indicates relatively well-defined material distribution in the fore region of the airfoil, from a manufacturing perspective, the resulting material distributions are infeasible and are attributed to relatively low aerodynamic suction and pressure forces acting on the airfoil at  $\alpha = 0$ .

In order to address this issue of infeasible material distribution within the airfoil structure, the adaptive compliant system is realized at a higher flow  $\alpha$ . A higher flow incidence in turn results in a relatively higher magnitude of suction and pressure force distribution over the OML, thus forcing the optimizer to distribute material in the aft and fore regions

to support the increased aerodynamic loads. Fig.8 represents increases in suction peak in the case of a NACA 0012 airfoil, at the fore section as the  $\alpha$  increases.

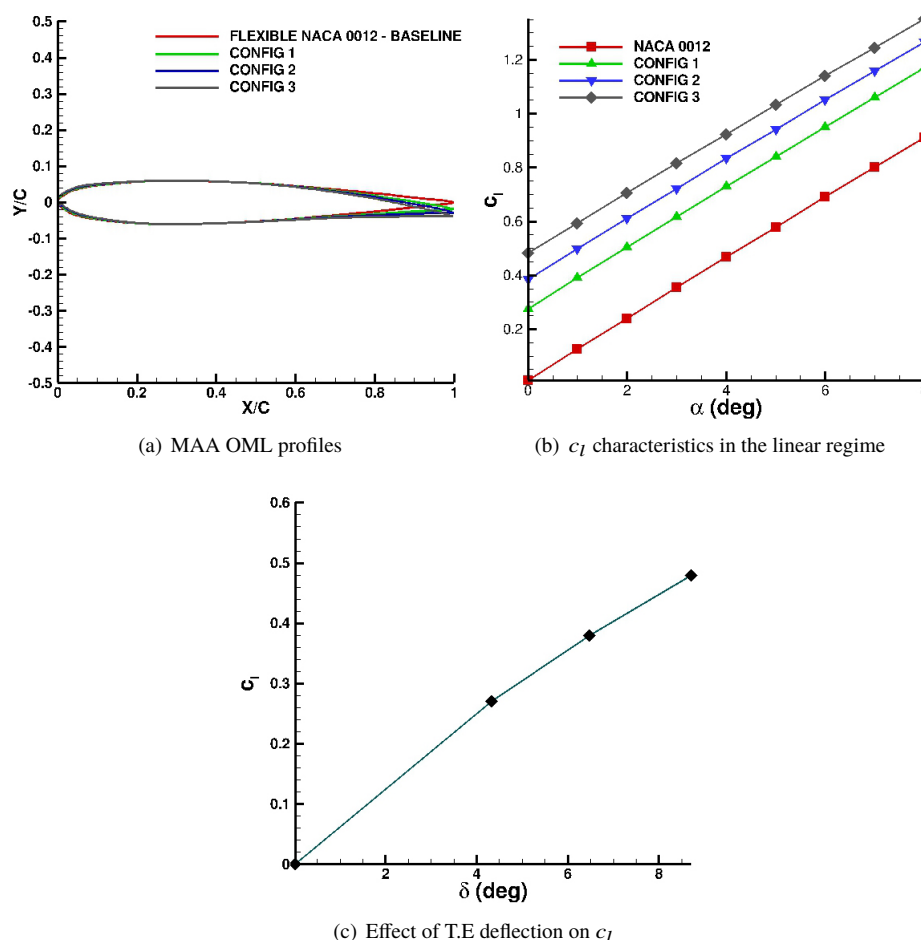


**Fig. 8 Normalized pressure distribution**



**Fig. 9 Adaptive airfoil configuration at  $\alpha = 2$**

Fig. 9 represents the internal mechanism layouts that constitute the MAA configuration, obtained using Eq. 21, when the structure is subjected to aerodynamic pressure loads at  $\alpha = 2^\circ$ . All  $c_l$  metrics in Fig. 9 are evaluated at  $\alpha = 0^\circ$ , and each of the four distinct configurations are obtained by solving the optimization problem subject to these design  $c_l$  requirements. The increase in suction and pressure forces at the OML results in a relatively well-defined load bearing structure in the aft and fore regions of the airfoil.



**Fig. 10 MAA configuration profile and performance trends**

Fig. 10(b) illustrates the gains in  $c_l$  from zero to moderate angles of attack. Clearly, the adaptive configurations produce the desired  $c_l$  characteristics at zero angle of attack. Such trends in  $c_l$  are similar to that of plain flap performance trends of the NACA 23012 wing section summarized in [28]. Fig. 10 suggests no change in  $c_{l\alpha}$  in the linear regime. Such trends are also intuitive since no fractional increments to the chord length  $c$  are made for either the plain flap or the MAA configurations. Fig. 10(c) represents the gains in  $c_l$  as a function of increasing T.E. deflection angle  $\delta$ . Similar trends for a NACA 23012 and a Clark Y airfoil with a plain flap are reported in [28].

In order to evaluate the performance characteristics of the mission adaptive configurations in a three-dimensional flow field, a series of RANS-based simulations were performed. Fig. 11 shows the  $c_d$  polar for the aeroelastically loaded NACA 0012 flexible airfoil, the MAA configuration and a multi-point design optimal configuration. Comparisons are made to a rigid NACA 0012 airfoil configuration. For low values of  $c_l$ , the rigid NACA 0012 airfoil is almost similar in terms of  $c_l - c_d$ . In fact, for  $c_l = 0$ , the flexible NACA 0012 configuration produces lesser drag than the rigid NACA 0012 configuration. However, for higher values of  $c_l$ , such similarities in trend diverge and the flexible NACA 0012 airfoil configuration under-performs. At the three design points ( $c_l^* = 0.27, 0.38, 0.48$ ), the MAA configuration is characterized by the least drag penalty. A multi-point optimal configuration as a result of unconstrained shape optimization yields the highest drag penalty. For  $c_l = 0$ , the MAA configuration is the flexible NACA 0012 with no actuation. The authors argue that such significant savings from the MAA configurations at the design point are mainly due to the elastic deformation of the OML due to aerodynamic pressure loads which in turn results in a OML profile which is significantly different from the rigid NACA 0012 airfoil.

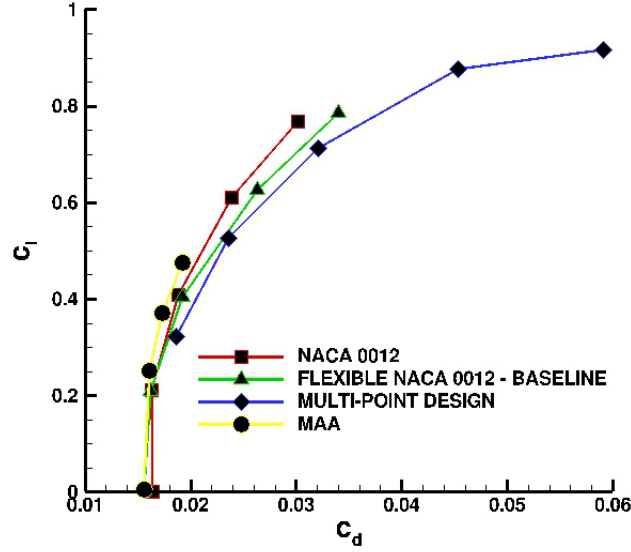


Fig. 11 Aerodynamic performance characteristics

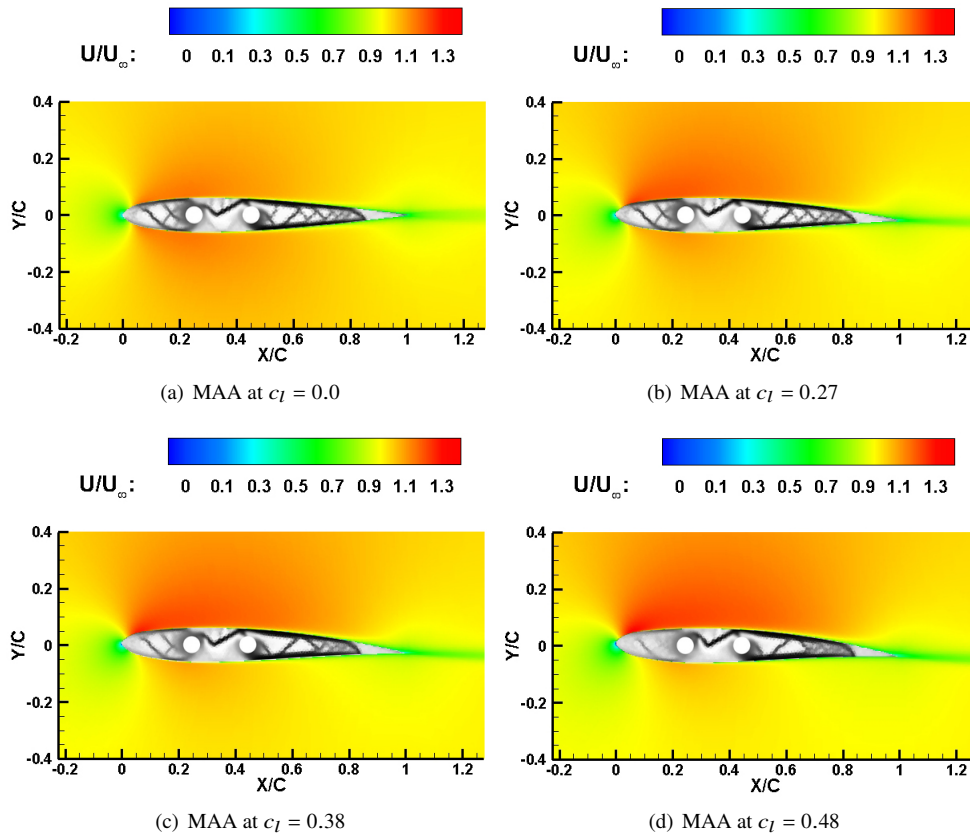
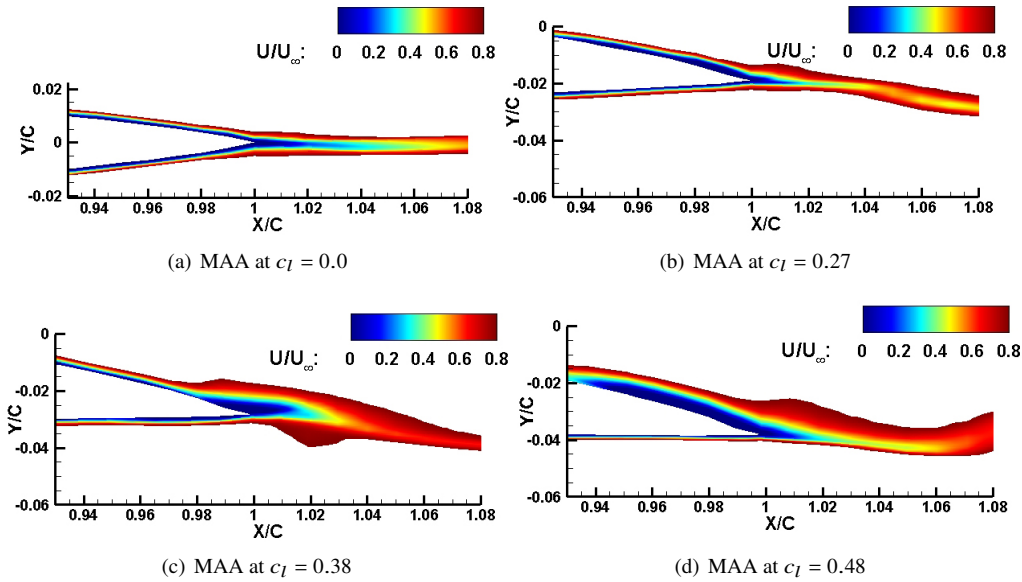


Fig. 12 Mean flow characteristics at design  $c_l$

Differences in OML due to external aerodynamic loading and internal actuation essentially results in flow field characteristics that are noticeably different from one another. Fig. 12 shows the mean flow field for all three MAA configurations and comparisons are drawn with the baseline flexible NACA 0012 airfoil configuration. Fig. 12(a)

suggests axisymmetric flow features especially in the regions close to the leading and trailing edges and the two suction peaks at the upper and lower surfaces. At  $c_l^* = 0.27$ , flow axisymmetry is lost even though the flow angle of attack  $\alpha$  is zero. Similar observations are made for the MAA configurations at  $c_l^* = 0.38$  and  $c_l^* = 0.48$ . The increases in the suction peak region on the upper surface as one progresses from Fig. 12(b) to (d) is attributed to the effect of increasing airfoil camber at fixed  $\alpha$  which results in higher lift. Instantaneous flow field characterization in the trailing edge region is shown in Fig. 13. The pair of axisymmetric trailing edge vortices in Fig. 13(a) is observed to coalesce to form a single vortex structure that tends to move towards the leading edge as  $c_l^*$  increases. The MAA configuration at  $c_l^* = 0.48$  is characterized by a significantly large re-circulation zone. One interesting feature of the MAA configurations in the trailing edge region is the extent of flow perturbation induced into the near field as the flow leaves the trailing edge. Such unsteady effects often form the foundation of trailing edge aero-acoustics studies. Evolving flow field features from Fig. 13 suggest that for high  $c_l^*$ , the separated boundary layer region may lead to increased viscous drag penalty and even sudden decrease of performance due to permanent boundary layer separation.



**Fig. 13** Instantaneous flow characteristics near trailing edge at design  $c_l$

## VII. Conclusions

A mission adaptive airfoil configuration was designed using an integrated multi-disciplinary design optimization framework. The MDO architecture was developed based on an aero-elastic system consisting of a linear finite element structural solver coupled with a constant strength doublet method for prediction of pressure forces over thick wing sections. The integrated optimization problem formulation leads to a downward deflection of the T.E and is subject to aerodynamic constraints on design  $c_l$ , mass fraction and structural compliance. Initial internal mechanism designs are characterized by intermediate designs which are further improved by subjecting the structure to higher pressure forces. The resulting MAA configurations are characterized by relatively well defined internal structure and favourable aerodynamic properties at the design points. Trends of  $c_{l,\alpha}$  suggest that the overall behavior of the MAA configurations are similar to airfoils with plain flap configurations. Near field flow characterization results in interesting evolving flow features as the  $c_l^*$  increases. Such effects are analogous to increasing the airfoil camber while keeping the  $\alpha$  fixed. The current study serves as a foundation for the design of Mission Adaptive Airfoil configurations with viscous drag considerations. The use of a non-linear finite element solver in place of a linear solver will allow the integrated framework developed in the current study to synthesize designs for higher  $c_l^*$ , thus resulting in high lift configurations.

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